Analyses: Modern Writing (TeDDi)

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Load libraries

If the libraries are not installed yet, you need to install them using, for example, the command: install.packages("ggplot2").

```
library(ggplot2)
library(ggrepel)
library(ggExtra)
library(gridExtra)
library(ggpubr)
```

Load Data

Load feature estimations with meta data for the Aurignacian mobile objects.

```
teddi.data <- read.csv("~/Github/UpperPalCombinatorics/results/teddi_features.csv")
```

Preselection

```
teddi.data <- teddi.data[teddi.data$num.char > 20 &
                           teddi.data$num.char < 200, ]
# select particular writing systems
# unique(teddi.data$writing_system)
teddi.data <- teddi.data[teddi.data$writing_system == "Arab"|
                           teddi.data$writing_system == "Geor"|
                           teddi.data$writing_system == "Hans"|
                           teddi.data$writing_system == "Thai", ]
```

TeDDi Specific Plots

Comparisons of different writing systems in TeDDi.

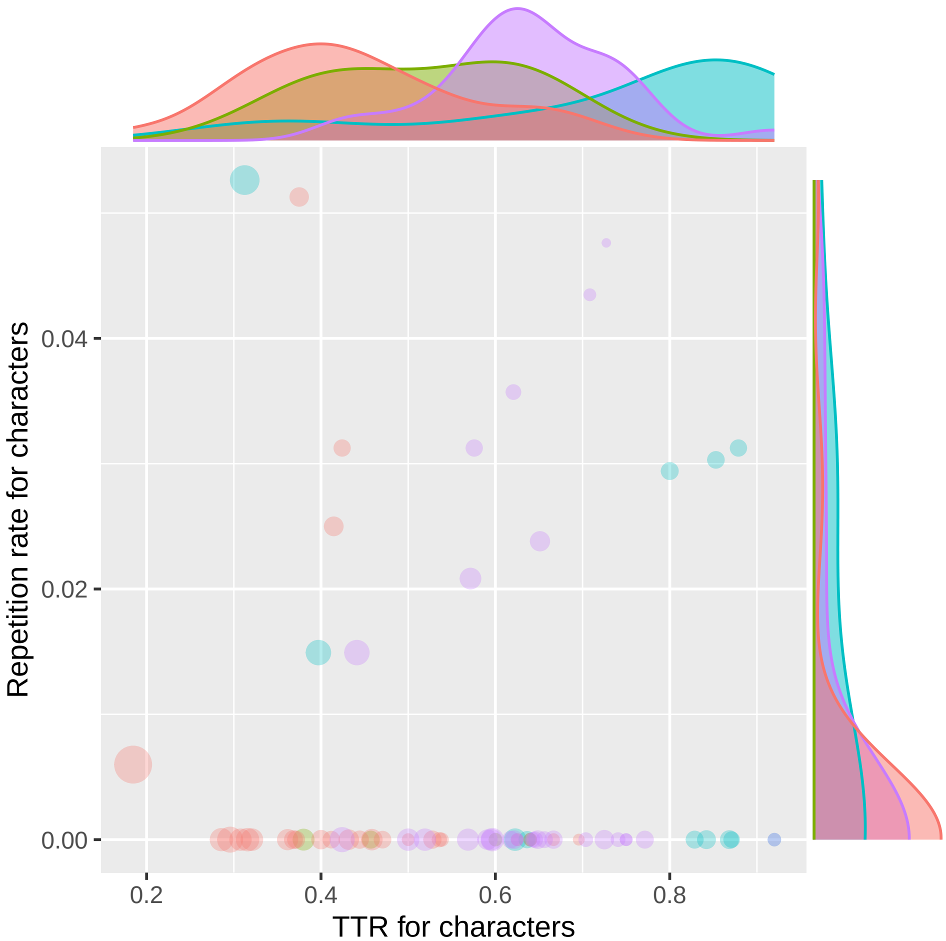
Entropy rate vs. unigram entropy

```
huni.hrate.chars.plot <- ggplot(teddi.data, aes(x = huni.chars, y = hrate.chars,
                                 colour = writing_system, size = num.char)) +
  geom_point(alpha = 0.3) +
  #geom_density2d(alpha = .8) +
  labs(x = "Unigram entropy for characters", y = "Entropy rate for characters") +
  theme(legend.position = c(.8,.2))
  #facet_wrap(~subcorpus)
huni.hrate.chars.plot <- ggMarginal(huni.hrate.chars.plot, groupFill = T, groupColour = T,
                                    type = "density")
huni.hrate.chars.plot
```

TTR vs. repetition rate

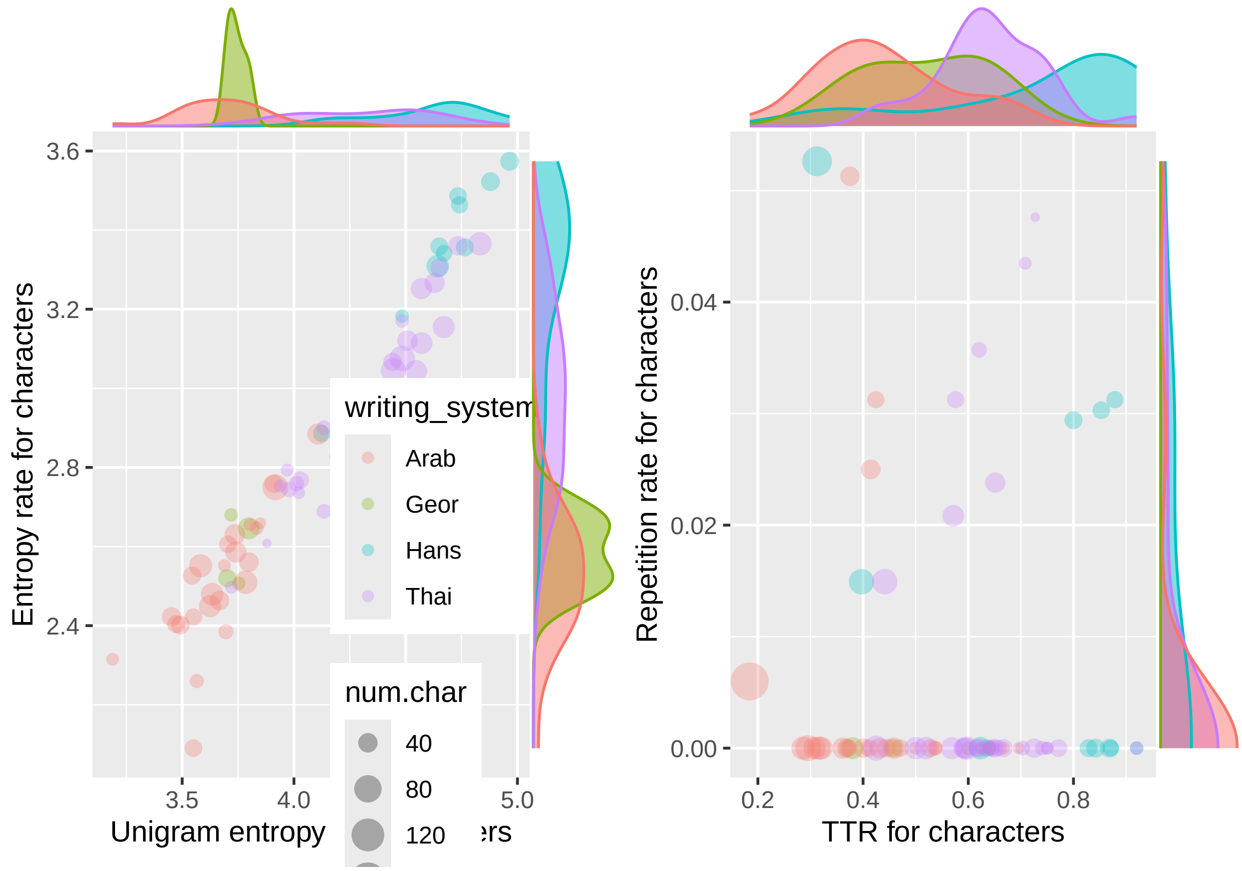
```
ttr.rm.chars.plot <- ggplot(teddi.data, aes(x = ttr.chars, y = rm.chars,
                                 colour = writing_system, size = num.char)) +
  geom_point(alpha = 0.3) +
  #geom_density2d(alpha = .5) +
  labs(x = "TTR for characters", y = "Repetition rate for characters") +
  theme(legend.position = "none")
```

```
  #facet_wrap(~subcorpus)
ttr.rm.chars.plot <- ggMarginal(ttr.rm.chars.plot, groupFill = T, groupColour = T,
                                type = "density")
ttr.rm.chars.plot
```



Combine figures.

```
combined.plot <- grid.arrange(huni.hrate.chars.plot, ttr.rm.chars.plot, ncol = 2)
```



Safe complete figure to file

```
ggsave("~/Github/UpperPalCombinatorics/figures/writingSystem_plot.pdf",
       combined.plot, dpi = 300, width = 12, height = 6, device = cairo_pdf)
```